

To complete the proof, our method will be to introduce a certain system of PDEs, then show that this system satisfies the Lopatinskii conditions [1], see also [44, Chapter 5, Proposition 11.9], if and only if λ is in the complement of the right side of (49). When the Lopatinskii conditions are satisfied, the system is a Fredholm operator which implies $M(\lambda)$ is also Fredholm. Therefore, in this case $\lambda \in \sigma_{ess}(L)^c$ which will establish the right inclusion of (49). The Lopatinskii conditions fail if either the system is not elliptic in the interior, or at the boundary. As we will see, interior ellipticity of the system is equivalent to

$$\lambda \in \left(\bigcup_{(x,\xi) \in M \times \mathbb{R}^3 \setminus \{0\}} \sigma_{pt}(x, \xi) \right)^c. \quad (53)$$

We have already shown that failure of this condition leads to existence of a Weyl sequence. Assuming interior ellipticity, we will show that boundary ellipticity is equivalent to

$$\lambda \in \left(\bigcup_{x \in \partial M} i|P_n \left[-\sqrt{\max(0, N^2)}, \sqrt{\max(0, N^2)} \right] \right)^c$$

We will show that failure of this condition also leads to existence of a Weyl sequence, which will complete the proof. Let us begin now deriving the PDE system.

For any $v \in H$ let us consider the decomposition given by Lemma 2, which can be written as

$$v = w + T^* \varphi$$

where $w \in \text{Ker}(T)$ and $\varphi \in H^1(M)$. Let us further decompose w according the standard Helmholtz decomposition as

$$w = \nabla \times (\rho_0 w_v) + \nabla \varphi_v$$

where $\varphi_v \in H^1(M)$ and the vector potential $\rho_0 w_v$ is in the space

$$H_{\text{Curl},0}(M) = \{u \in L^2(\rho_0 \, dx) : \nabla \times u \in L^2(\rho_0 \, dx), \, n \times u|_{\partial M} = 0\},$$

while also satisfying

$$\nabla \cdot (\rho_0 w_v) = 0.$$

Given that M is a ball, a unique such decomposition exists (see [2, Section 3]). Let us set $\rho_0 z_v = \nabla \varphi_v$ which must then satisfy

$$\nabla \times (\rho_0 z_v) = 0.$$

Then $w \in \text{Ker}(T)$ is equivalent to

$$\nabla \cdot (\rho_0 z_v) + \frac{g'_0}{c^2} \cdot \nabla \times (\rho_0 w_v) + \frac{\rho_0 g'_0}{c^2} \cdot z_v = 0, \quad n \cdot z_v|_{\partial M} = 0.$$

Now, suppose that $u \in \text{Ker}(T)$ satisfies

$$M(\lambda)u = f. \quad (54)$$

As described above for v , there will be w_u and z_u such that

$$u = \nabla \times (\rho_0 w_u) + \rho_0 z_u$$

where

$$\nabla \times (\rho_0 z_u) = 0, \quad (55)$$

$$\nabla \cdot (\rho_0 w_u) = 0, \quad (56)$$

$$\nabla \cdot (\rho_0 z_u) + \frac{g'_0}{c^2} \cdot \nabla \times (\rho_0 w_u) + \frac{\rho_0 g'_0}{c^2} \cdot z_u = 0, \quad (57)$$

$$n \cdot z_u|_{\partial M} = 0, \quad (58)$$

$$n \times w_u|_{\partial M} = 0. \quad (59)$$